

PanoramaScore Agentic Risk Pipeline Proposal

Submitted by: Asad Nawaz Date: May 22, 2025

■ Project Overview

PanoramaScore is a multi-agent AI pipeline that automates risk scoring across different verticals using document understanding, NLP, F

The pipeline follows an 8-step agentic workflow:

1. Document Ingest Agent: Accepts PDF/image/CSV files
2. Metadata Hashing Agent: Records file source, ID, checksum
3. OCR & Normalization Agent: Extracts and cleans text
4. NLU Extraction Agent: Identifies structured fields from text
5. Vector Store & RAG Agent: Embeds data and retrieves similar cases
6. Domain Adapter Agent: Maps generic fields to vertical-specific risk schema
7. Risk Scoring Engine Agent: Outputs numeric score, band, and factor contributions
8. Copilot Delivery Agent: Sends JSON response with highlights and action buttons

✓ Selected Verticals:

- Insurance
- Real Estate
- Supply Chain
- Fraud Detection

✗ Skipped: Cross-Asset Investing (due to high complexity & less immediate benefit)

--- Milestones Breakdown ---

■ Milestone 1 (Week 1): Setup + Ingestion + Metadata

Tools: FastAPI, Pydantic, hashlib

Tasks: File upload, store document, extract checksum, setup project structure

■ Milestone 2 (Week 2): OCR + Field Extraction

Tools: PyMuPDF, Tesseract, SpaCy, Regex

Tasks: Extract text, clean it, extract key fields, confidence scores

■ Milestone 3 (Week 3): RAG + Vector Store

Tools: sentence-transformers, pgvector, LangGraph

Tasks: Create embeddings, store & retrieve similar past cases

■ Milestone 4 (Week 4): YAML Domain Adapter + Risk Scoring

Tools: PyYAML, LightGBM/CatBoost

Tasks: YAML mapping, scoring logic, threshold bands

■ Milestone 5 (Weeks 5–6): Delivery Agent + Testing + Docs

Tools: LangGraph, FastAPI, Swagger/OpenAPI

Tasks: Final integration, audit logs, documentation

Each step will be reviewed before moving forward. Quality code and modular structure are mandatory.